

12.159

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Question

The determinant

$$\begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (0.1)$$

evaluates to

- a) 0
- b) $2b(b-1)$
- c) $2(1-b)(1+2b)$
- d) $3b(1+b)$

Theoretical solution

Let the determinant of the given matrix be Δ . We will find its value by converting the matrix to Row Echelon Form (REF).

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b & 1+b & 1 \\ 1 & 2b & 1 \end{vmatrix} \quad (0.2)$$

First, apply the operation $R_2 \rightarrow R_2 - R_1$.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ b-(1+b) & 1+b-b & 1-1 \\ 1 & 2b & 1 \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 1 & 2b & 1 \end{vmatrix} \quad (0.3)$$

Theoretical solution

Next, apply the operation $R_3 \rightarrow R_3 - R_1$.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 1-(1+b) & 2b-b & 1-1 \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ -b & b & 0 \end{vmatrix} \quad (0.4)$$

Here, we can observe that the third row is a scalar multiple of the second row: $R_3 = b \cdot R_2$. This implies the rows are linearly dependent.

Theoretical solution

To complete the reduction to REF, we perform one final operation, $R_3 \rightarrow R_3 - bR_2$, to create a zero row.

$$\Delta = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ -b-b(-1) & b-b(1) & 0-b(0) \end{vmatrix} = \begin{vmatrix} 1+b & b & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix} \quad (0.5)$$

Since the matrix in Row Echelon Form has a row of all zeros, its determinant is 0.

$$\Delta = 0 \quad (0.6)$$

This corresponds to option (a).